SUPPLEMENTAL MATERIAL

Data S1. Newcastle-Ottawa scoring

NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE - CASE CONTROL STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability.

Selection

1) Is the case definition adequate?

a) yes, with independent validation *

b) yes, e.g. record linkage or based on self-reports

c) no description

2) Representativeness of the cases

a) consecutive or obviously representative series of cases *

b) potential for selection biases or not stated

3) Selection of Controls

a) community controls *

b) hospital controls

c) no description

4) Definition of Controls

a) no history of disease (endpoint) *

b) no description of source

Comparability

1) Comparability of cases and controls on the basis of the design or analysis

a) study controls for CVD risk factors (Select the most important factor.) *

b) study controls for any additional factor * (This criterion could be modified to indicate specific control for a second important factor.)

Exposure

1) Ascertainment of exposure

a) secure record (e.g., surgical records) *

b) structured interview where blind to case/control status *

c) interview not blinded to case/control status

d) written self-report or medical record only

e) no description

2) Same method of ascertainment for cases and controls

a) yes *

b) no

3) Non-Response rate

a) same rate for both groups *

b) non respondents described

c) rate different and no designation

NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE - COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort

a) truly representative of the average population (e.g., general population or CKD patients) of that age in the community *

b) somewhat representative of the average population (e.g., general population or CKD patients) of that age in the community *

c) selected group of users e.g., nurses, volunteers

d) no description of the derivation of the cohort

2) Selection of the non-exposed cohort

a) drawn from the same community as the exposed cohort *

b) drawn from a different source

c) no description of the derivation of the non-exposed cohort

3) Ascertainment of exposure

a) secure record (e.g., surgical records) *

b) structured interview *

c) written self-report

d) no description

4) Demonstration that outcome of interest was not present at start of study

a) yes *

b) no

Comparability

1) Comparability of cohorts on the basis of the design or analysis

a) study controls for previous history of cardiovascular disease *

b) study controls for any additional conventional cardiovascular risk factors *

Outcome

1) Assessment of outcome

a) independent blind assessment *

b) record linkage *

c) self report

d) no description

2) Was follow-up long enough for outcomes to occur

a) yes (select an adequate follow up period for outcome of interest) *

b) no

3) Adequacy of follow up of cohorts

a) complete follow up - all subjects accounted for *

b) subjects lost to follow up unlikely to introduce bias - small number lost - > 20% lost to follow up, or description provided of those lost) *

c) follow up rate < 20% and no description of those lost

d) no statement

Results

All studies provided a clear description of how exposure (amount of AAC) was ascertained, however there was significant variability in between-study test characteristics and methods of measurement. Scores used to determine AAC varied between the 52 studies, 19 studies (37%) used thresholds based on the Kauppila AAC 8 or 24 score, 14 studies (27%) used the presence vs. the absence of AAC, 7 studies (13%) used thresholds based on the AAC index (ACI), 5 studies (10%) used thresholds based on Agatston score and the remaining 7 studies (13%) used other scoring methods. Only 15 studies (30%) provided evidence that prevalent cardiovascular disease was not present at the beginning of the study. The majority of studies (94%) adjusted for conventional cardiovascular risk factors or additional disease-specific risk factors. However, only 11 studies (21%) provided adjustment for the history of cardiovascular disease. Only 25 studies (48%) reported a complete/near complete follow-up.

Table S1. Search strategy.

Keyword	MEDLINE	Embase
Population = None	No search strategy	No search strategy
Intervention/Test = Abdominal aortic calcification	exp Vascular Calcification/ or exp Calcinosis/ or exp Vascular Diseases/ or arterial calcification.mp. or exp Arteriosclerosis/ or exp Arterial Occlusive Diseases/ or exp Aortic Diseases/ or aortic.mp. or vascular calcifications.mp. or exp Vascular Calcification/ and abdomin$.mp. and aortic calc$.mp.	vascular calcification.mp. or exp blood vessel calcification/ or artery calcification.mp. or exp artery calcification/ or exp coronary artery disease/ or exp aorta atherosclerosis/ or exp aorta disease/ or exp arteriosclerosis/ or arteriosclerosis.mp. or exp atherosclerosis/ or exp atherosclerotic plaque/ or extracoronary.mp. and abdomin$.mp. and artery calc$.mp.
Methodology = observational	prognosis.sh. or diagnosed.tw. or cohort$.mp. or predictor$.tw. or death.tw. or exp models, statistical	prognosis.sh. or diagnosed.tw. or cohort$.mp. or predictor$.tw. or death.tw. or exp models, statistical
Comparator = None	No search strategy	No search strategy
Outcome = None	No search strategy	No search strategy
Additional specific filters	Human	Human

***The reference lists of recent literature reviews and guidelines were hand-searched for further studies.

Table S2. Detailed GRADE assessment for outcomes in people recruited from the general population*.

	Quality of assessment (Decrease in quality score)				Effect size/dose response	Summary of findings		
Studies† (subjects)	Risk of bias	Inconsistency	Indirectness	Imprecision		Relative risk (95% CI)	Test for heterogeneity	Quality of evidence
Cardiovascular events								
6 (8,498)	-	↓1 between-study heterogeneity	↓ popn. selection may not be generalizable	no serious imprecision	0/↑1	1.83 (1.40-2.39)	I^2=87%, p-value<0.001	Moderate
Fatal cardiovascular events								
5 (8,004)	-	↓1 between-study heterogeneity	↓ popn. selection may not be generalizable	no serious imprecision	0/↑1	1.85 (1.44-2.39)	I^2=69%, p-value=0.001	Moderate
All-cause mortality								
5 (8,862)	evidence of publication bias (↓1)	↓1 between-study heterogeneity	↓ popn. selection may not be generalizable	no serious imprecision	↑1/ ↑1	1.98 (1.55-2.53)	I^2=90%, p-value<0.001	Moderate
Coronary heart disease events								
5 (7,766)	-	↓1 between-study heterogeneity	↓ popn. selection may not be generalizable	no serious imprecision	↑1/ ↑1	2.22 (1.57-3.15)	I^2=72%, p-value<0.001	High
Cerebrovascular events								
4 (8,943)	-	↓1 between-study heterogeneity	↓ popn. selection may not be generalizable	no serious imprecision	↑1/ ↑1	2.10 (1.41-3.12)	I^2=60%, p-value<0.001	High

***** All outcomes are considered clinically important. Scores are based on the GRADE for assessment of evidence about prognosis where the evidence begins as high quality evidence.[18] These criteria are based on; a) 5 domains diminishing confidence (-1 for risk of bias, inconsistency, imprecision, indirectness, and publication bias) and b) 2 situations increasing confidence (+1 or +2 for large (RR >2) to very large (RR >4) effect size and a +1 for a dose-response gradient [increasing pooled relative risks for CV events, fatal CV events, CHD events, cerebrovascular events and all-cause mortality with increasing severity of AAC]).[18]
† Number of studies with suitable data for meta-analysis.

Table S3. Detailed GRADE assessment for outcomes in chronic kidney disease patients*

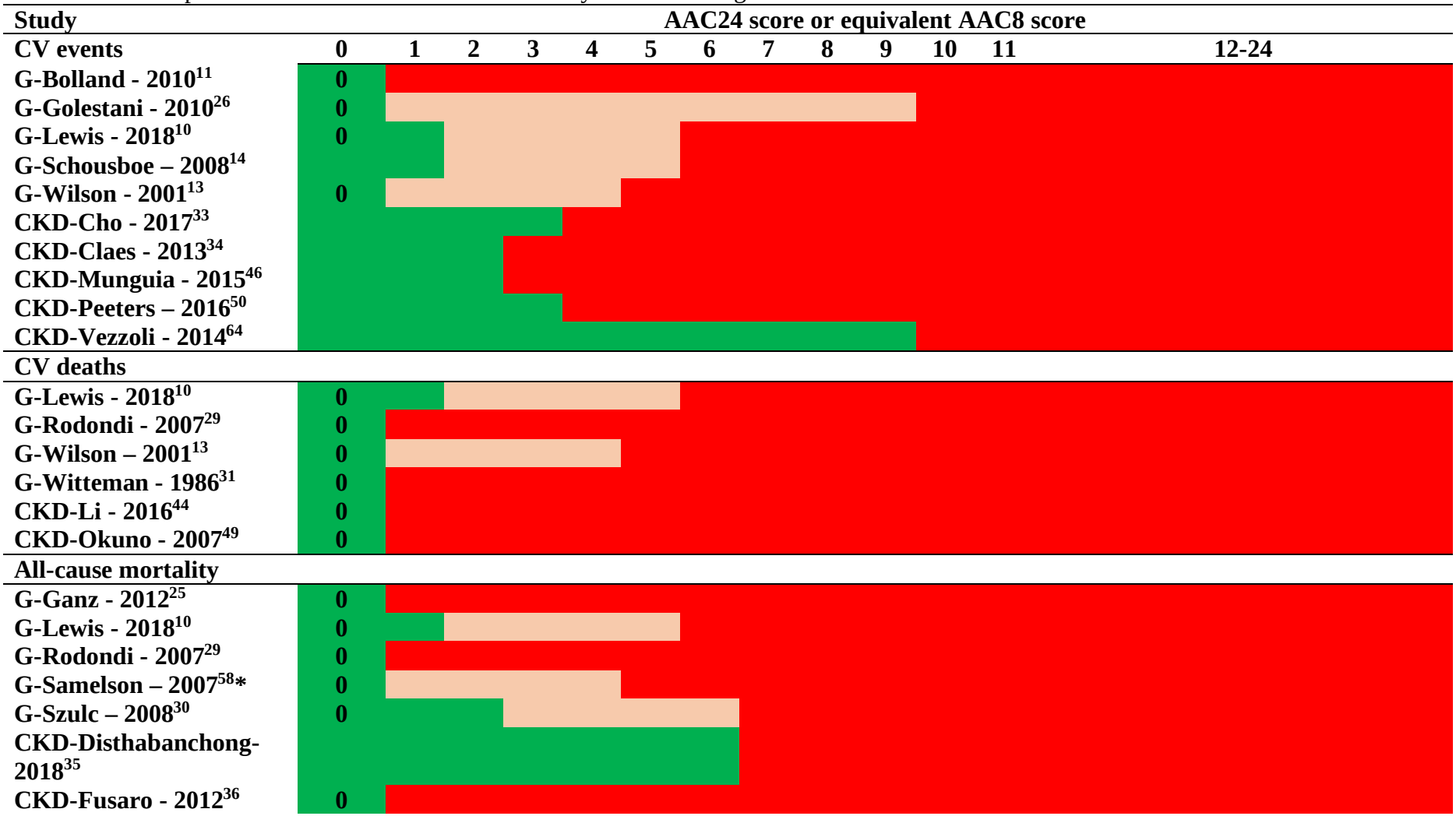
	Quality of assessment (Decrease in quality score)				Effect size/dose response	Summary of findings		
Studies† (subjects)	Risk of bias	Inconsistency	Indirectness	Imprecision		Relative risk (95% CI)	Test for heterogeneity	Quality of evidence
Cardiovascular events								
8 (1,426)	evidence of publication bias (↓1)	Low/moderate between-study heterogeneity	popn. generalizable	no serious imprecision	↑1/ insufficient evidence‡	3.47 (2.21, 5.45)	I^2=29%, p-value=0.196	Moderate
Fatal cardiovascular events								
4 (1,163)	-	Low between-study heterogeneity	popn. generalizable	no serious imprecision	↑1/ insufficient evidence‡	3.68 (2.32, 5.84)	I^2=0%, p-value=0.788	High
All-cause mortality								
9 (2,050)	-	Low between-study heterogeneity	popn. generalizable	no serious imprecision	↑1/ insufficient evidence‡	2.40 (1.95-2.97)	I^2=0%, p-value=0.584	High

***** All outcomes are considered clinically important. Scores are based on the GRADE for assessment of evidence about prognosis where the evidence begins as high quality evidence.[18] These criteria are based on; a) 5 domains diminishing confidence (-1 for risk of bias, inconsistency, imprecision, indirectness, and publication bias) and b) 2 situations increasing confidence (+1 or +2 for large (RR >2) to very large (RR >4) effect size and a +1 for a dose-response gradient [increasing pooled relative risks for CV events and all-cause mortality with increasing severity of AAC]).

† Number of studies with suitable data for meta-analysis

‡Less than 3 studies reported three or more groups.

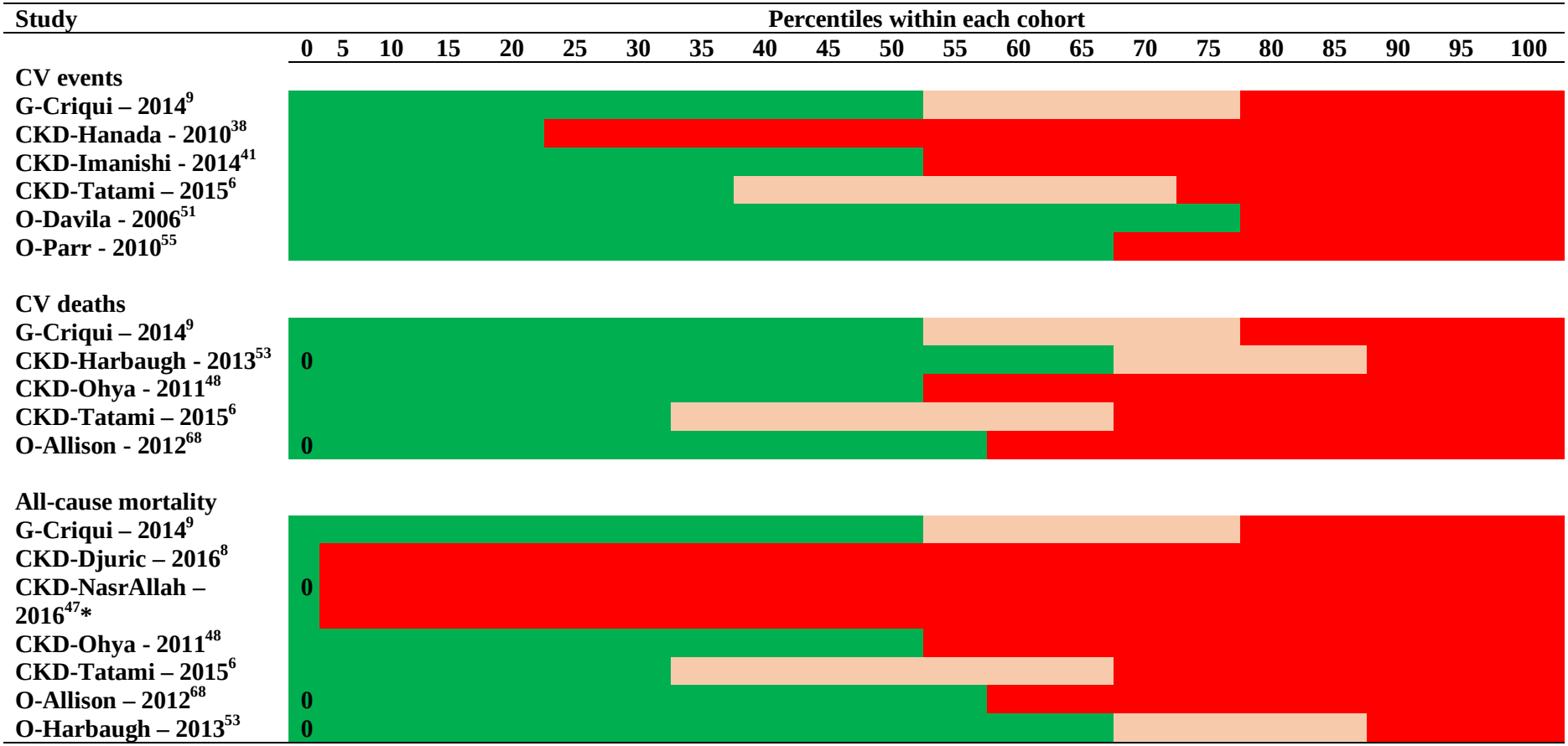
Table S4. The reported AAC cut-offs for studies of X-rays or DXA images.



CKD-Hong - 2013[39]	**0**
CKD-Li - 2016[44]	**0**
CKD-Munguia - 2015[46]	
CKD-NasrAllah - 2017[47]	**0**
CKD-Okuno - 2007[49]	**0**

G=recruited from the general population, CKD=recruited from the chronic kidney disease population. Green indicates the lowest reported category of AAC, pink indicates the moderate AAC and red indicates the highest reported category of AAC.
*No extractable data for all-cause mortality in three groups.

Table S5. The reported AAC cut-offs for studies using CT.



G=recruited from the general population, O, recruited form other populations, CKD=recruited from the chronic kidney disease population. Zero indicated categorised according to the presence of AAC. Green indicates the lowest reported category of AAC, pink indicates the moderate AAC and red indicates the highest reported category of AAC. * No extractable data for some outcomes.

Table S6. Multivariable adjustments used for individual studies of pooled adjusted estimated of risk.

Study reference	Point estimates (HR or RR) adjusted for
General population	
Allison 2012[68]	age, sex, body mass index, smoking, diabetes, hypertension, dyslipidemia, and family history of CVD.
Bolland 2010[11]	age, systolic BP, smoking, status, history of diabetes, history of blood pressure treatment and BMI for women or total cholesterol for men
Golestani 2010[26]	age, gender, hypertension, hypercholesterolemia, diabetes mellitus, smoking, and family history of CHD.
Hollander 2003[28]	Adjusted for age, sex, diabetes mellitus, smoking, systolic and diastolic BP, cholesterol and HDL cholesterol, and history of CVD.
Lewis 2018[10]	Framingham risk model (using BMI) and treatment code
Levitzky 2008[57]	age, gender, diabetes, systolic BP, left ventricular hypertrophy on electrocardiogram, BMI, total cholesterol, HDL cholesterol, current cigarette smoking, and hypertension treatment.
Rodondi 2007[29]	age, smoking status, diabetes, systolic blood pressure, physical activity, waist girth, and history of angina and myocardial infarction.
Schousboe 2008[14]	age, systolic BP, LDL and HDL cholesterol, triglycerides, smoking, renal function, treatment assignment (clodronate or placebo), self-reported diagnoses of diabetes mellitus, hypertension, angina, prior stroke, and health status.
Szulc 2008[30]	age, weight, tobacco smoking, diabetes, and medications.
Samelson 2007[58]	age, BMI, smoking, systolic BP, total cholesterol, diabetes, CHD, and estrogen use (in women).
Wilson 2001[13]	age, cigarettes, diabetes mellitus, systolic BP, left ventricular hypertrophy, BMI, cholesterol, and HDL cholesterol
Witteman 1986[31]	Blood pressure, total serum cholesterol, BMI, smoking history (current, past, never) and diabetes

Table S7. Comparison of random and fixed effect estimates of studies from the general population.

Any advanced abdominal aortic calcification	**Random effects RR (95% CI)**	**Fixed effects RR (95% CI)**
Chronic kidney diease		
Cardiovascular events	3.47 (2.21, 5.45)	3.30 (2.29-4.77)
Fatal cardiovascular events	3.69 (2.32-5.85)	3.69 (2.32-5.85)
All-cause mortality	2.41 (1.95-2.97)	2.41 (1.95-2.97)
General population		
Cardiovascular events	1.83 (1.40-2.39)	1.56 (1.45-1.68)
Fatal cardiovascular events	1.85 (1.44-2.39)	1.89 (1.68-2.13)
All-cause mortality	1.98 (1.56-2.93)	1.49 (1.42-1.55)
Coronary heart disease	2.23 (1.57-3.15)	1.98 (1.76-2.28)
Cerebrovascular disease	2.10 (1.41-3.12)	2.11 (1.70-2.62)

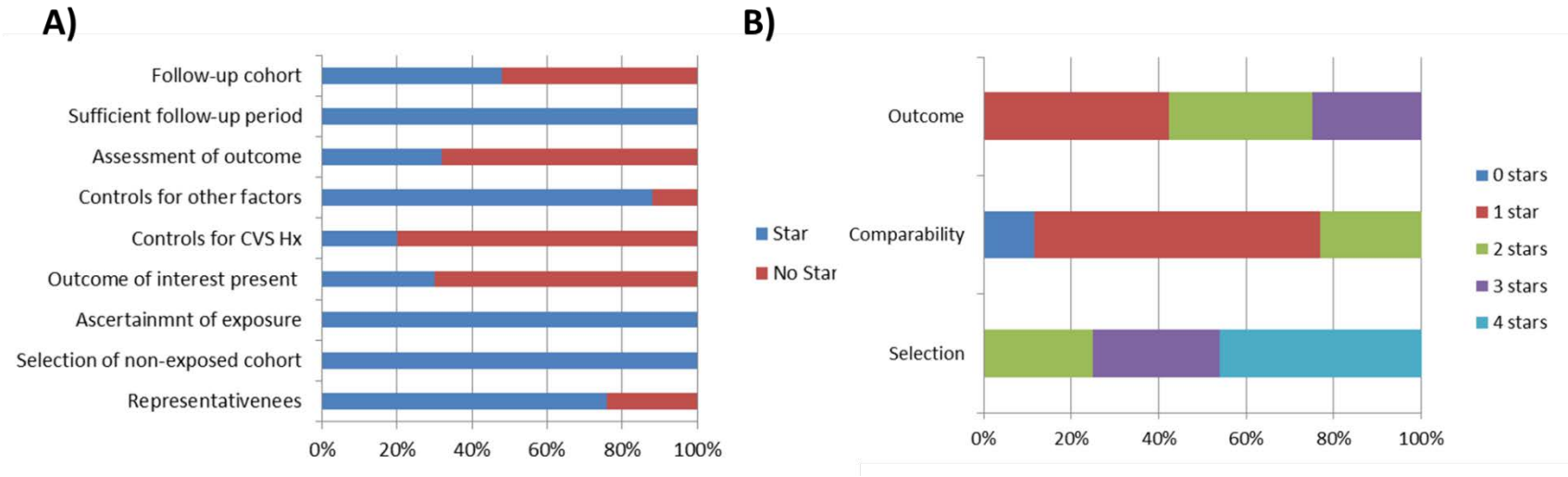


Figure S1. Newcastle Ottawa scoring of studies included in the meta-analysis by a) items and b) domains. Maximum number of stars for each domain is 1) outcome maximum of 3 stars 2) comparability maximum of 2 stars, and 3) Selection maximum of 4 stars. CVS Hx = History of cardiovascular disease which was considered the most important factor to be controlled for.

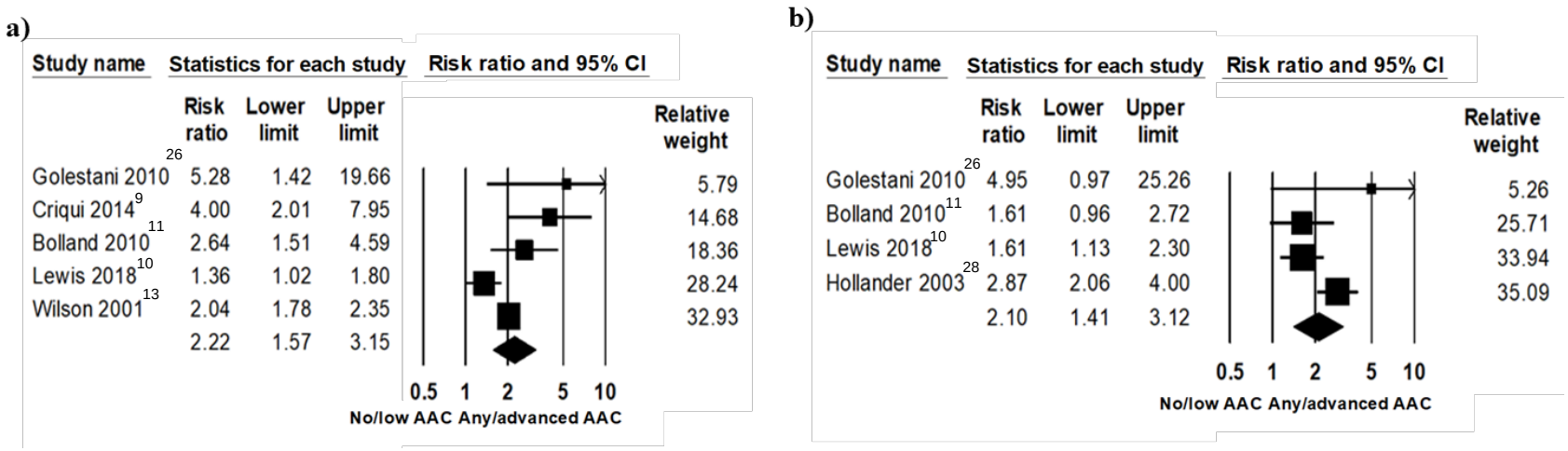


Figure S2. Forest plots for the pooled relative risk of a) coronary heart disease events (I^2=72%, p-value=0.012) and b) cerebrovascular events (I^2=60%, p-value=0.055) in the general population stratified by no or low abdominal aortic calcification (AAC) vs any or more advanced AAC.

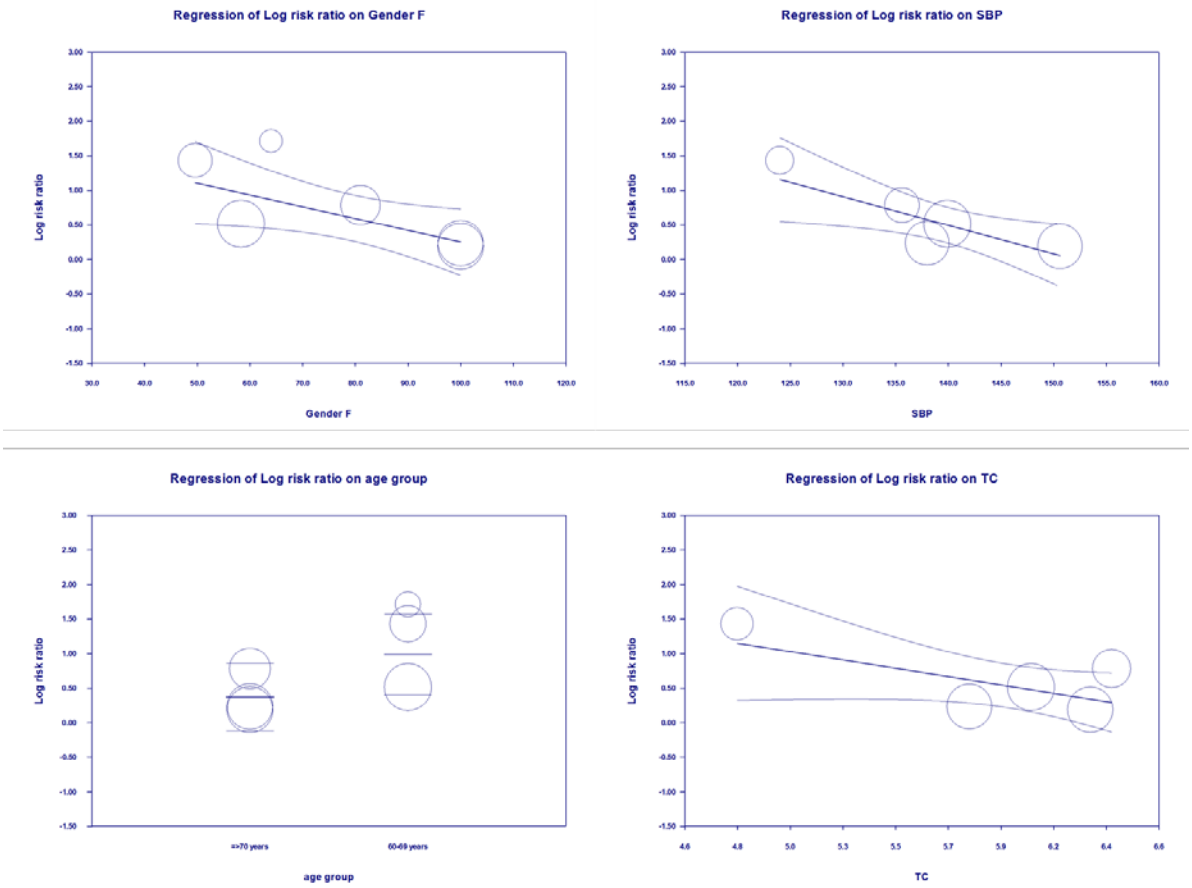


Figure S3. Meta-regression – Log RR of any CV event by top left) the proportion of women the study cohort (r^2=0.0, p=0.012), top right), baseline systolic blood pressure (SBP) in mmHg (r^2=0.42, p=0.003), bottom left) age group (r^2=0.0, p=0.048) and bottom right) total cholesterol in mg/dL (r^2=0.04, p=0.048). Variables tested (where sufficient studies were available n=3) included: age group, total cholesterol, high density lipoprotein cholesterol, % females, %type 2 diabetics, & current smokers, BMI, SBP, % prescribed anti-hypertensives, % with clinical history of CVD.

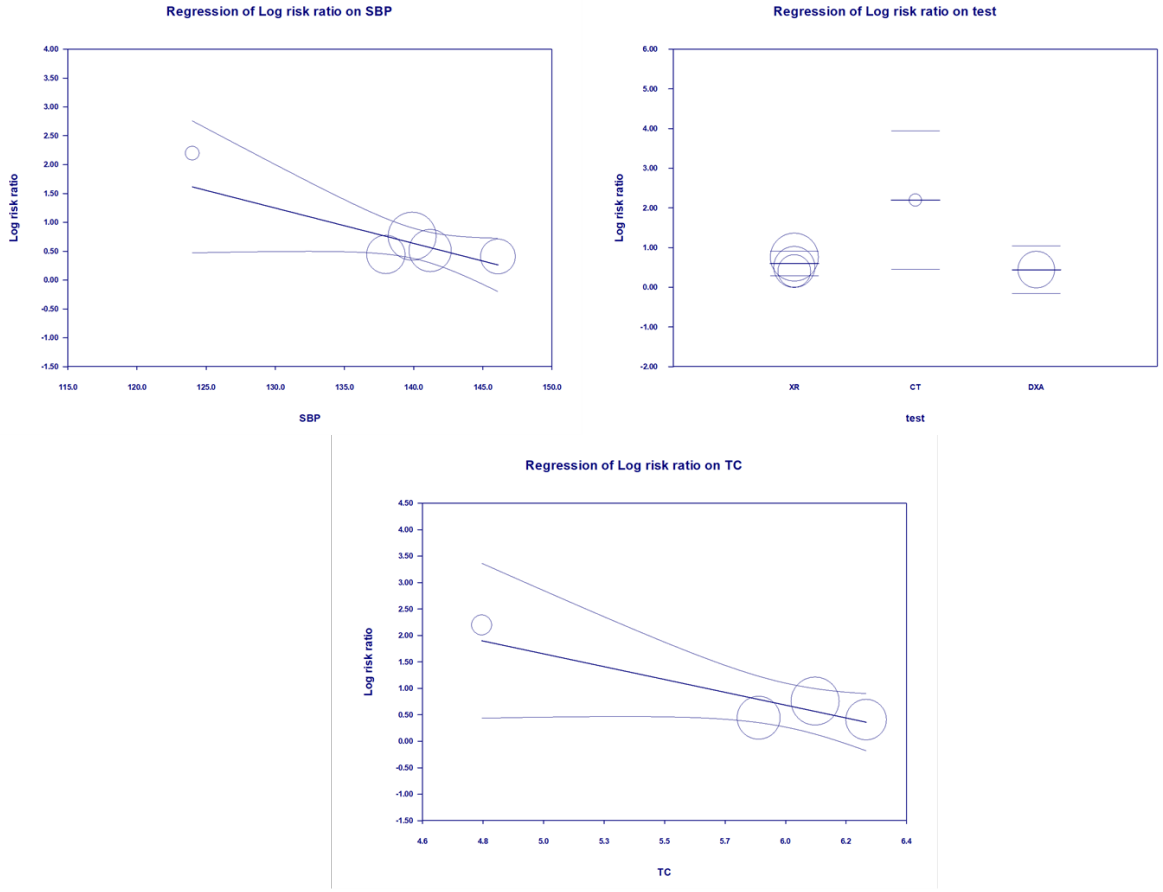


Figure S4. Meta-regression – Log RR for fatal CV events top left) imaging modality used in the studies (r^2=0.60, p=0.029) and top right) means systolic blood pressure (SBP) in mmHg (r^2=0.45, p=0.027) and bottom left) total cholesterol in mg/dL (r^2=0.13, p=0.031). Variables tested (where sufficient studies were available n=3) included age group, total cholesterol, high density lipoprotein cholesterol, % females, %type 2 diabetics, & current smokers, BMI, SBP, % prescribed anti-hypertensives, % with clinical history of CVD.

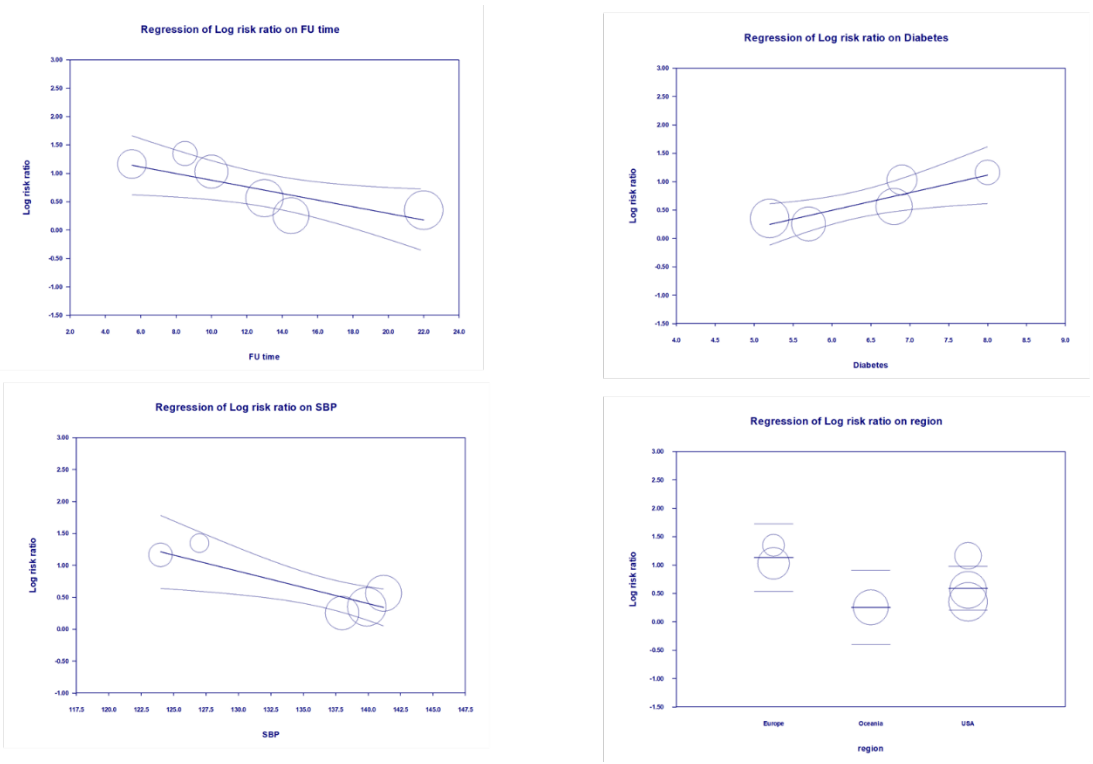


Figure S5. Meta-regression – Log RR of any all-cause mortality by top left) follow up time in year (r^2=0.11, p=0.008) and top right) % of participants with type 2 diabetes (r^2=0.42, p=0.003), bottom left) mean systolic blood pressure (SBP) in mmHg (r^2=0.39, p=0.002) and bottom right) region of study (r^2=0.36, p=0.017). Variables tested (where sufficient studies were available n=3) included age group, total cholesterol, high density lipoprotein cholesterol, %females, %type 2 diabetics, & current smokers, BMI, SBP, % prescribed anti-hypertensives, % with clinical history of CVD.